Module: CH401 – Dance of the Electron
Topic: Chemical Thermodynamics
Subtopic: Entropy and Free Energy
School: National Junior College
Class: JH402
Class size: 27
Class tutor: Mr Harman Johll
Stream: Junior High
Students' ability: High
Environment: Classroom
Number of periods: Two 30 min periods

Prerequisite Knowledge

Before the commencement of this module, students are expected to be able to:

1. Describe the meaning of enthalpy change in terms of exothermic and endothermic reactions
2. Describe exothermic and endothermic reactions in terms of heat change and its effect on the temperature of the reaction mixture and/or its surroundings
3. Represent energy changes by energy profile diagrams, including reaction enthalpy changes and activtaion energies
4. Explain and use the term entropy (i.e. simply state that it is a measure of (dis)order).

Specific Instructional Objectives (Overarching and Standards' Measure)

At the end of the lesson package for this week, students should be able to:

1. explain the term entropy
2. predict whether the entropy change for a given process/reaction is positive or negative by considering the effects of changes in temperature, phase and mixing on the entropy
3. state and use the equation $dG = dH - TdS$
4. understand that the spontaneity of a chemical reaction and/or physical process is determined by the sign of dG, i.e. a balance of the change in enthalpy and entropy of the system for a given temperature
5. predict the effect of temperature change on the spontaneity of a reaction, given standard enthalpy and entropy changes

Instructional Objectives Differentiated For This Class

1. To recap the definition of entropy, viz. state that it is a measure of disorder (e.g. gas has higher entropy than solid, a greater number of particles might imply a greater entropy)
2. Predict whether the entropy change for a given process/reaction is positive or negative by considering the phases and particle numbers of reactants and products
3. Understand that both enthalpy and entropy are required to predict if a reaction is spontaneous
4. State the 2nd Law of Thermodynamics, derive the equation dG = dH –TdS, and make meaning of the equation
5. Understand that spontaneity is determined by dG (or $dS_{univ} > 0$)
6. Apply 2nd Law of Thermodynamics, the relationship between dG and equilibrium position, and the equation dG = dH – TdS to increase yields

Teaching Aids (Activities and Materials)

Two in-class demonstrations (see Lesson Development for details) will be used to teach this topic to this group of students:

1. Solid-solid reaction that results in formation of ice (beaker sticks to block or mat, and latter is lifted)
2. Imploding soft drink bottle (with lots of heat)

One set of props, viz. shoelace and ring, will be used to illustrate how yields of a reaction are related to dG.

A worksheet/activity sheet which students will use to follow the lesson, and the whiteboard is all that is required.

Motivation For Lesson

1. Students should understand the meaning of entropy and how it applies

2. Students should be learn to be comfortable with deriving equations, specifically how the equation dG = dH – TdS can be derived from the statement of the Second Law of Thermodynamics

3. Students should understand the meaning of equations and not simply use them

4. Students should take understand that these ideas form the crux of *why things happen the way they do*.

Key Learning Points

1. Entropy measures the degree of randomness or disorder in a system.
2. Spontaneity of the process is favoured by an increase in entropy of a system (dS > 0)
3. The sign of free energy change, dG, is directly related to reaction spontaneity; a negative dG corresponds to a spontaneous reaction
4. dG depends on both enthalpy and entropy changes of a process, and is related by dG = dH – TdS
5. Temperature change affects the spontaneity of a reaction

General Summary from Topic of Chemical Thermodynamics (points to reference during lessons)

1. The spontaneity of a process (physical or chemical) can be predicted using the change in Gibbs' Free Energy (dG). A spontaneous process has a negative dG.
2. dG is dependent on both dH and dS
3. dH is the change in enthalpy of a reaction, brought about by a change in energy of the reacting substances. This is often manifested in the form of heat transfer, causing temperature changes during a reaction.
4. However, direct measurement of heat change due to a reaction is not always possible. dH of a reaction can be more easily determined by making use of Hess' Law.
5. dH can be calculated by using bond energy values. The enthalpy change for a reaction is equal to the bond energy of bonds broken – bond energy of bonds formed.
6. Another way to calculate dH is by making use of enthalpy change of formation values, which provides a practical way to combine thermochemical equations of known 'step' reactions.
7. dS is the change in degree of randomness or disorder in a system. It is closely linked to statistical probability, because an event that brings about disorder is more likely to occur than one that brings about order.
8. Although a reaction may be thermodynamically feasible, it might not be seen to take place because it is very slow (limited by kinetics).

Lesson Development

Time (min)	Activity	Remarks
5 (5)	Get class settled. Distribute handouts and worksheets to class. This is the first lesson with this class this term. Re-iterate classroom expectations and highlight some of the key learning points they will be taking away this term. Remind them of the year-end assessment. While class chemistry representative helps, write down the aims (in question form) of the lesson. Point out to students that these aims are not typical outcomes expected but is something that every chemist **must absolutely** understand and appreciate. The following should be written and left on the board throughout the class. 1. What is the meaning of entropy and what is the use of knowing entropy? (Is knowledge of enthalpy sufficient?) 2. dG tells us what is useful work. What else did we learn (from re-arranging the expression for dG in terms of dH and dS)? 3. Why do things happen and can we apply equations meaningfully?	Get class chemistry representative to help with this. Motivate the main reasons for studying chemical thermodynamics: it tells us why stuff happens, and it tells us what we should do when we carry out a particular reaction (e.g. organic synthesis). Reference organic reactions carried out at MCL research exchange programme where liquid nitrogen is used to cool the reacting system in particular cases where adducts are formed.

Time (min)	Activity	Remarks
3 (8)	Review students' understanding of entropy by getting them to state if the reactants or the products of 5 specific reactions have higher or lower entropy. Sound/call out 5 different students to vocalize the answer. If time permits (depending on length of assembly since this is the first lesson of term), it would be prudent to get students to state the type of reactions referenced here.	

Time (min)	Activity	Remarks
5 (13)	Demonstration I: The Imploding Bottle Prepare the system: throw in some Alka-Seltzer tablets and 'pump' the system with carbon dioxide gas. Give a running commentary of what is being done. At the same time, ask students to think about why this process seems so favourable, and something we hardly even think about or consider. Also get students to re-cap if the environment should feel warm (cold) for exothermic (endothermic) reactions. 'Throw in' the sodium hydroxide solution. Get a student to vocalize the change that occurred. Do not simply accept a response which describes what happened, e.g. "the bottle caved in" or "the bottle imploded" or "the gas went away". Prompt students to state the gas evolved reacted with or perhaps dissolved in the sodium hydroxide solution that was added; the gas might be acidic. Ask students if there is a decrease in entropy of the system. Get one student to feel the bottle and ask him/her to state if it is warm to the touch. Get another student to state if the reaction is exothermic or endothermic.	

Time (min)	Activity	Remarks
5 (18)	Demonstration II: Solid and Solid Reaction Ask students if they would expect an endothermic or exothermic would occur more readily or with more spontaneity. In the context of enthalpy change, there is a decrease in enthalpy (energy) during an exothermic process. In some sense, there is added stability. Now demonstrate the discrepant event: a very spontaneous endothermic (extremely endothermic) reaction. While carrying out demonstration, give running commentary and get students to fill out the necessary items on page 2 of their worksheet: 2. How did the consistency change upon mixing the two solids together 3. Was there any gas evolved? What gas was evolved? 4. What special “heavy lifting” could the beaker do? Why could it do this? 5. Write the chemical reaction for this reaction. State if the entropy increased or decreased on going from reactants to products.	

Time (min)	Activity	Remarks
7 (30)	The Second Law of Thermodynamics – The Crux of It All – Why Stuff Happens? Share with students that the 2nd Law of Thermodynamics is personal favorite. No matter what changes there might be in understanding nature, this law will surely stand the test of time. It is a law loaded with information even though it is expressed so simply by an inequality. So what then is entropy? First, define entropy based on SIO requirement, viz. a measure of disorder of the system. Get students to then think about what drives disorder: e.g. heat, temperature, state, probability, permutations, etc. are possible answers. Lead students to arrive at the two variables that define entropy: heat and temperature. **Figure 1 of Worksheet, The Crux of The Dance** Supply thermal energy. The more <u>heat</u> or <u>thermal energy</u> supplied, the <u>higher</u> the entropy. It seems like the change in entropy is <u>directly</u> proportional to the <u>heat/thermal energy</u> supplied. Question students: is it fair to simply consider the thermal energy supplied? Think carefully about what we are really concerned with here: **disorder**. Is the thermal energy supplied in all scenarios going to cause the same extent of disorder? Give students 3-5 minutes to draw the change in organisation of particles on going from 1pK to 10 K and 3000K to 3010 K for any species. It seems like that change in entropy is <u>indirectly</u> proportional to the temperature at which the chemical or physical process occurs.	

Time (min)	Activity	Remarks
10 (40)	**Playing with equations and giving mathematical expressions physical meaning** Re-iterate to students that knowing formula is one thing, knowing how nature works is another. To do this, it is important to dissect and derive meaning from equations. Derive, starting with the idea that the universe is made up of the system and its surroundings the following: $dS_{univ} = dS_{sys} + dS_{surr}$ $dS_{univ} = \frac{dQ_{sys}}{T} + \frac{dQ_{surr}}{T}$ $TdS_{univ} = dQ_{sys} + dQ_{surr}$ $TdS_{univ} = dQ_{sys} - dH_{sys}$ $-TdS_{univ} = TdS - dH$ $dG = dH - TdS$ The critical point in the above derivation rests on understanding that the heat change associated with the surrounding is the **negative of the heat change associated with the system, viz. the negative of the enthalpy change of the system**. *Possible question that might be brought up by students: if it is the enthalpy of the system is the negative of the heat change of the surroundings, then should the expression not yield zero? This question is perfectly valid if and only if the process or change was done with thermodynamic reversibility. Any difference accounts for unrecovered heat loss.* The next critical step is for students to do a simple re-arrangement of the equation. Give students a minute to try this re-arrangement on their own and ask them to shout out what it might mean.	

Time (min)	Activity	Remarks
10 (50)	Now explain the use of dG. State explicitly and get students to note phrasing: If dG is negative, the process is spontaneous; if dG is positive, the process is not spontaneous. Explain to students that dG is a thermodynamic measure and informs us of the yield we might expect from a chemical or physical process. It tells us for example that ice will not form at 298K, or rather has a very low probability of forming at that temperature. It also tells us that diamond should turn in graphite. The value of dG however says nothing about how fast a reaction or process will proceed. In other words, kinetic information is missing from the Gibbs energy value. dG tells us what to expect at the end of a process. It tells us yield, and the equation tells us what we can do to improve that yield. There will always be reactants and products at the *end* of a process. The question is how much. The greater the difference in Gibbs energy between reactant and product, the greater the amount of products formed at the end of the process. If dG is an intermediate or small value, a certain percentage or ratio of reactants or products will be established, and this is only changed by changing the temperature (dG = dH – TdS). Refer students to worksheet page 5. Are there reactants and products at the end of the reaction? <u>Yes</u>	Assign homework. Get students to name the isomers that they have come up with from the previous exercise.

Time (min)	Activity	Remarks
Cont'd	However, the extent to which reactants turn to products depends on how large the magnitude of dG is. If the products have a much lower G than the reactants, then the end state would be predominantly products. If the products have a G value that is just slightly higher or lower than the reactants, then the end state would favour a little more reactants and products respectively.	
3 (53)	Application to synthesis. Run through questions 8(a) and 8(b) *ad verbatim* with the class. For question 8(a), ask Victoria to share her MCL reaction with the class. For question 8(b), ask Milla to consider the desorption case of her research project.	
3 (56)	Get students to try in pairs, to prove that thermal energy cannot 'flow' from a cold body to a hot body.	
4 (60)	Conclusion. Re-iterate points stated in aim of lesson (i.e. those mentioned at the start of class). Make use of items written on board. 1. What is the meaning of entropy and what is the use of knowing entropy? (Is knowledge of enthalpy sufficient?) 2. dG tells us what is useful work. What else did we learn (from re-arranging the expression for dG in terms of dH and dS)? 3. Why do things happen and can we apply the equations confidently? Practice will be given to apply the equation. This is straightforward and trivial. The most important or critical idea to take away is knowing how to apply this when carrying out reactions to understand how to manipulate the temperature to get as much yield as possible, while at the same time taking into account the kinetics of the reaction.	

Lesson Reflection